Mini Project

Exploratory Data Analysis on India's COVID-19 Data

State-wise trends, recovery rates, and vaccination progress — with a fulldbde walkthrough and written findings

Companion project to Module 1: Python for Data Science

Data sources: *imdevskp/covid-19-india-data* (case data) and *Our World in Data* (vaccination data)

Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn

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Report prepared for postgraduate and BCA Data Science training programs

Contents

1 Objective 2

2 Step-by-Step Workflow (How-To Guide) 2
- 2.1 Step 1 — Load the data . . . 2
- 2.2 Step 2 — Inspect before touching anything . . . 2
- 2.3 Step 3 — Clean . . . 3
- 2.4 Step 4 — Explore with groupby and pivot_table . . . 3
- 2.5 Step 5 — Visualise and Step 6 — Summarise . . . 3

3 Charts and Findings 4
- 3.1 National trends over time . . . 4
- 3.2 Which states were hit hardest? . . . 5
- 3.3 Recovery and death rates . . . 6
- 3.4 How metrics relate to each other, and how the outbreak evolved month by month . . . 8
- 3.5 The other end of the spectrum, and the vaccination follow-up . . . 9

4 Overall Summary 10
- 4.1 Caveats . . . 10
- 4.2 Suggestions for further analysis . . . 10

1 Objective

The goal of this mini project is simple: take a real, publicly available, slightly messy dataset about India's COVID-19 outbreak, and turn it into a clear picture of what happened — state by state, over time — using exactly the tools covered in the Module 1 handbook: Pandas, NumPy, and Matplotlib/Seaborn.

Two datasets are used together:

- Case data — daily, state-wise confirmed cases, deaths, and recoveries, sourced from India's Ministry of Health (MoHFW) and covid19india.org bulletins, covering 30 January 2020 to 6 August 2020 (India's first wave).
- Vaccination data — India's national daily vaccination totals from Our World in Data, covering January 2021 onward, once the vaccine rollout began.

> **Note**
> These two datasets don't overlap in time — the case data ends before vaccines existed. We treat them as two chapters of the same story: how the outbreak spread, and how the public health response scaled up afterward.

2 Step-by-Step Workflow (How-To Guide)

This section walks through exactly how the analysis was built, in order, so you can reproduce it yourself in the companion notebook (miniproject_covid_eda.ipynb).

2.1 Step 1 — Load the data

Both datasets are read directly from their public GitHub/OWID URLs with pandas.read_csv() — no manual download required.

```
import pandas as pd

cases = pd.read_csv(
    "https://raw.githubusercontent.com/imdevskp/"
    "covid-19-india-data/master/complete.csv"
)
vax = pd.read_csv(
    "https://raw.githubusercontent.com/owid/covid-19-data/master/"
    "public/data/vaccinations/country_data/India.csv"
)
```

2.2 Step 2 — Inspect before touching anything

```
cases.dtypes
cases.isna().sum()
cases["Name of State / UT"].nunique()
```

This inspection immediately reveals two real-world data problems:

1. The Death column is stored as text, not a number.

2. The same state is spelled multiple ways — e.g. Telangana, Telengana, and Telangana*** are all one state, and some Union Territories appear twice (once plain, once prefixed with "Union Territory of ...").

2.3 Step 3 — Clean

```
cases.columns = [
    "date", "state", "lat", "long", "confirmed",
    "deaths", "cured", "new_cases", "new_deaths", "new_recovered",
]
cases["date"] = pd.to_datetime(cases["date"])
cases["deaths"] = pd.to_numeric(cases["deaths"], errors="coerce").fillna(0).astype(int
    )

name_fix = {
    "Telangana***": "Telangana",
    "Telengana": "Telangana",
    "Union Territory of Jammu and Kashmir": "Jammu and Kashmir",
    "Union Territory of Ladakh": "Ladakh",
    "Union Territory of Chandigarh": "Chandigarh",
}
cases["state"] = cases["state"].replace(name_fix)
cases = cases.drop_duplicates(subset=["date", "state"])
cases["active"] = cases["confirmed"] - cases["deaths"] - cases["cured"]
```

Key Idea

Clean once, at the start, before any grouping or plotting. If you fix a spelling mistake *after* you've already run a groupby("state"), every chart before that point is silently wrong.

2.4 Step 4 — Explore with groupby and pivot_table

```
# National daily totals
national = cases.groupby("date", as_index=False)[
    ["confirmed", "deaths", "cured"]
].sum()
national["recovery_rate"] = (national["cured"] / national["confirmed"] * 100).round(2)

# Latest snapshot per state, sorted
latest = cases[cases["date"] == cases["date"].max()]
latest = latest.sort_values("confirmed", ascending=False)

# Case growth by state and month
cases["month"] = cases["date"].dt.strftime("%Y-%m")
pivot = cases.pivot_table(
    values="new_cases", index="state", columns="month", aggfunc="sum", fill_value=0
)
```

2.5 Step 5 — Visualise and Step 6 — Summarise

The remaining sections of this report *are* Steps 5 and 6: twelve charts built with Matplotlib and Seaborn, followed by a written summary of what they show.

3 Charts and Findings

3.1 National trends over time

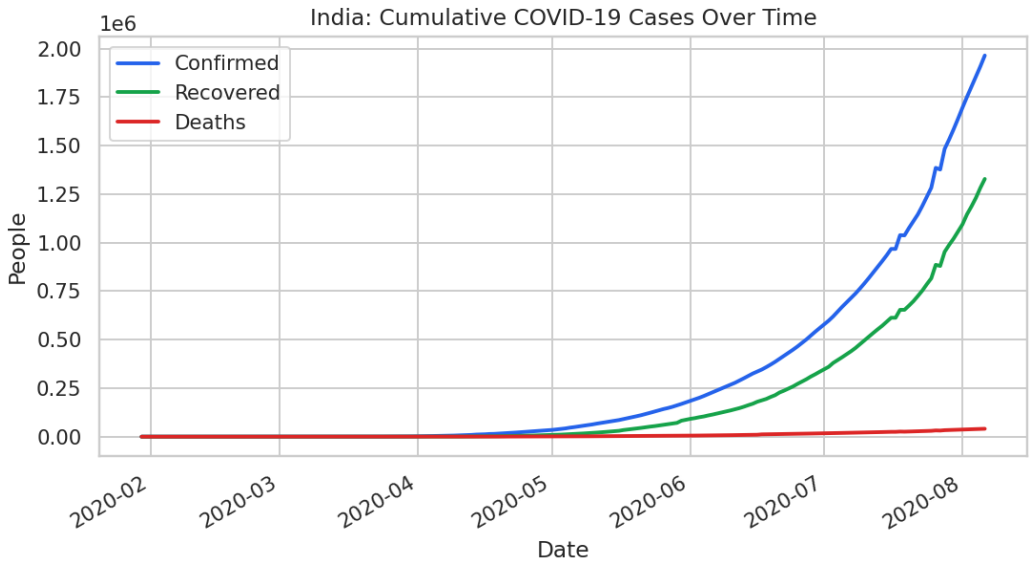


Figure 1: Cumulative confirmed cases, recoveries, and deaths, nationally.

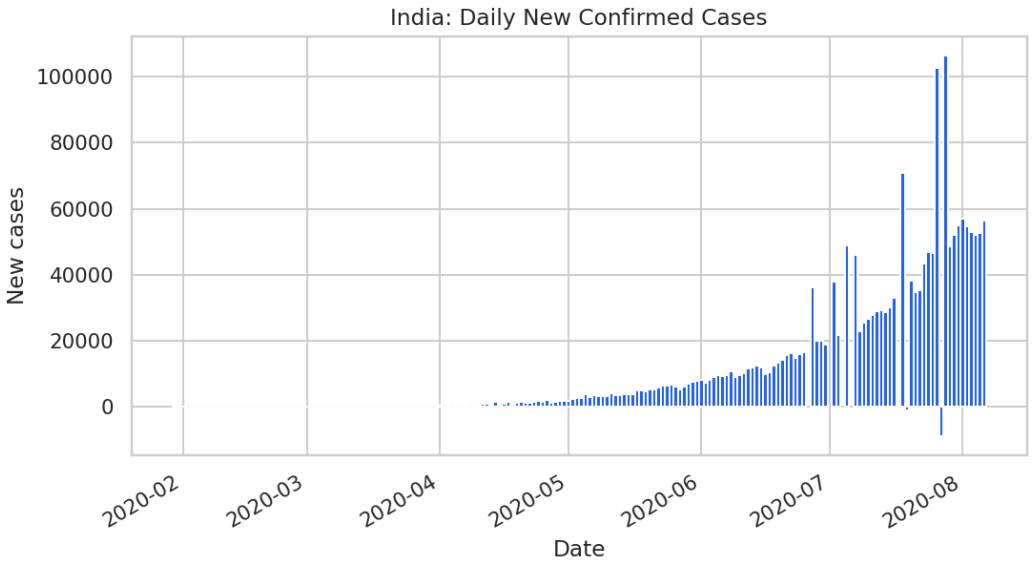


Figure 2: Daily new confirmed cases, nationally.

Finding

By 6 August 2020, India had recorded about 19.6 lakh (1.96 million) confirmed cases, 13.3 lakh recoveries, and roughly 40,700 deaths. Crucially, the daily new-case count was still rising sharply at the end of the data window — it peaked around 1,06,000 new cases in a single day in late July 2020 — which tells us the first wave had not yet crested by this point.

3.2 Which states were hit hardest?

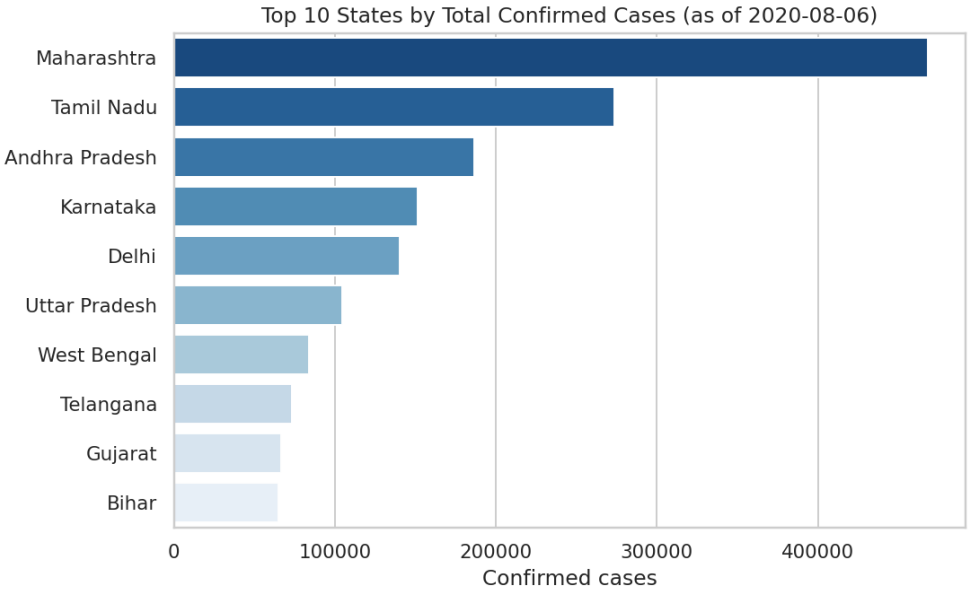


Figure 3: Top 10 states by total confirmed cases.

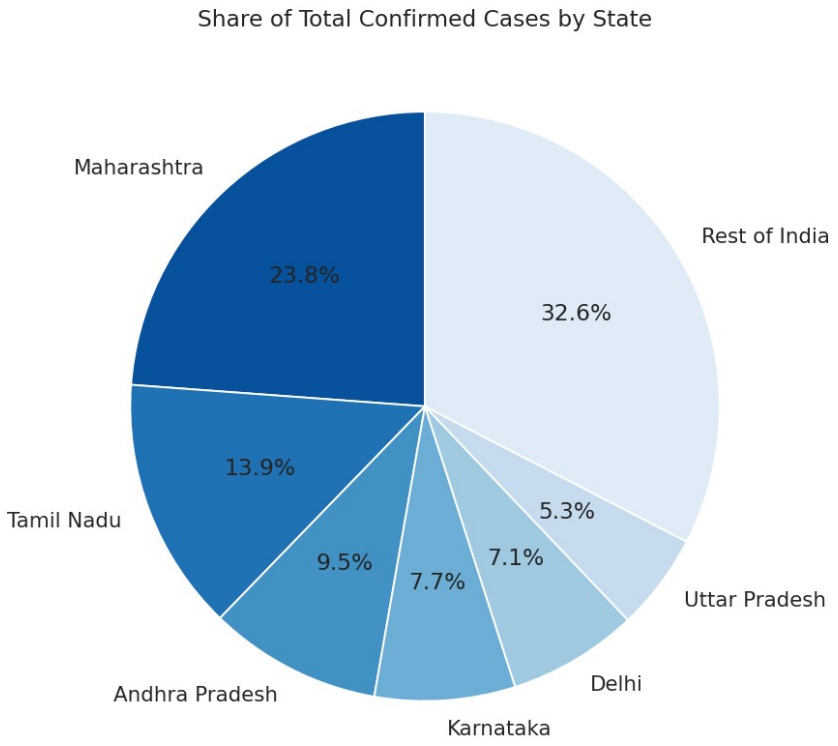


Figure 4: Share of national confirmed cases held by the top states.

Finding

The outbreak was heavily concentrated, not evenly spread. Maharashtra alone accounted for close to a quarter of all national cases (~4.68 lakh), with Tamil Nadu, Andhra Pradesh, Karnataka, and Delhi rounding out the top five. Together, the top six states held the clear majority of the country's total case load.

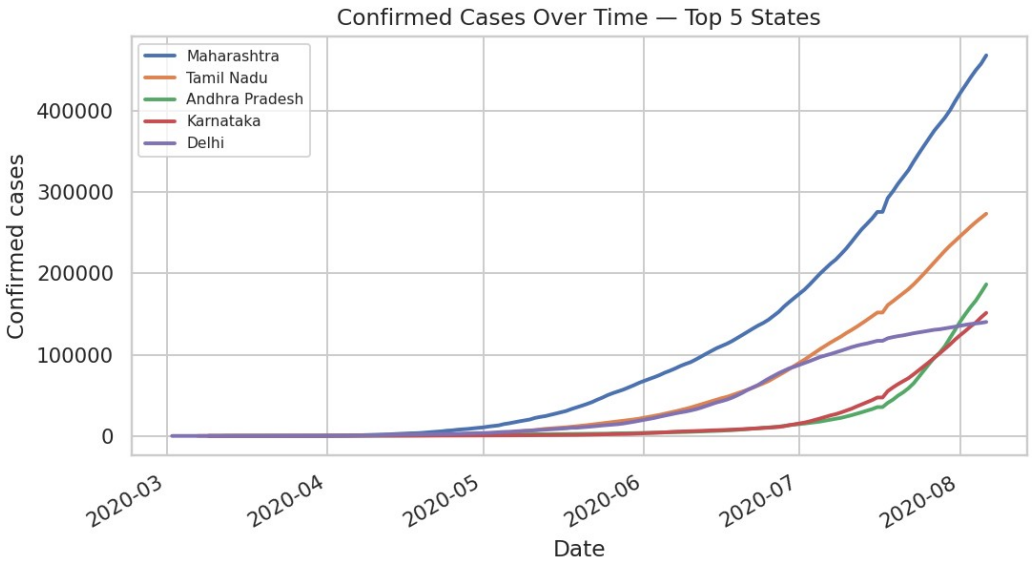


Figure 5: Confirmed-case growth curves for the top 5 states, over time.

3.3 Recovery and death rates

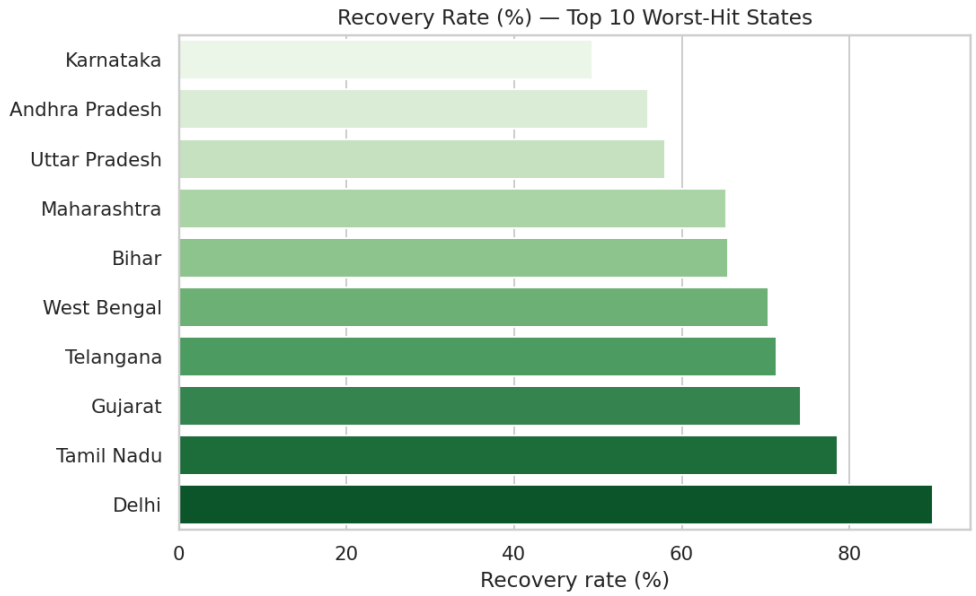


Figure 6: Recovery rate (%) for the ten worst-hit states.

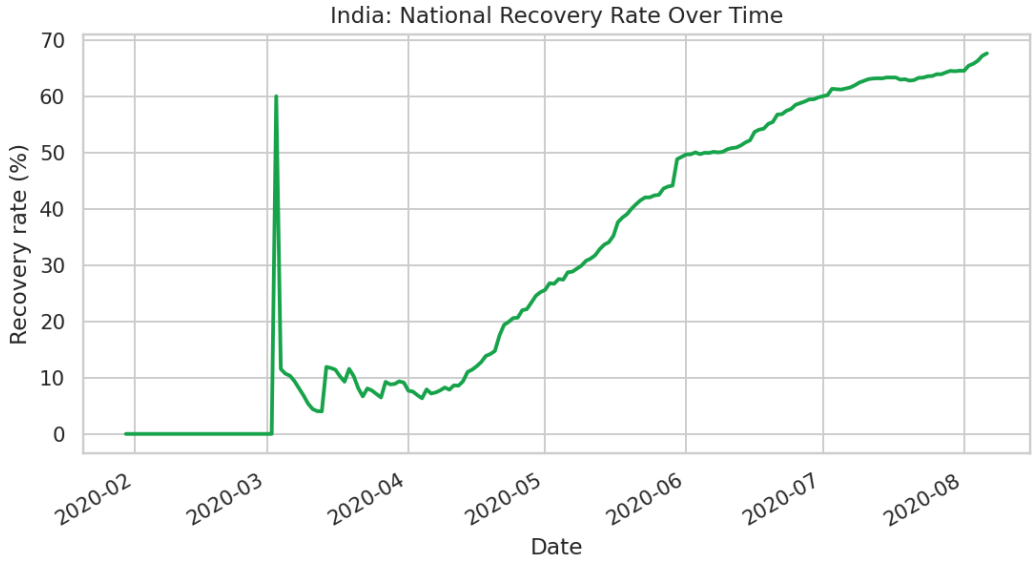


Figure 7: National recovery rate over time.

Finding

Recovery varied a lot state to stateDelhi stands out with a recovery rate near 90% — well ahead of every other high-case state — while Karnataka was still under 50% recovered at this snapshot, suggesting Karnataka's outbreak was comparatively newer or faster-growing relative to how much time had passed for patients to recoverNationally, the recovery rate climbed steadily throughout the window, ending near 68%.

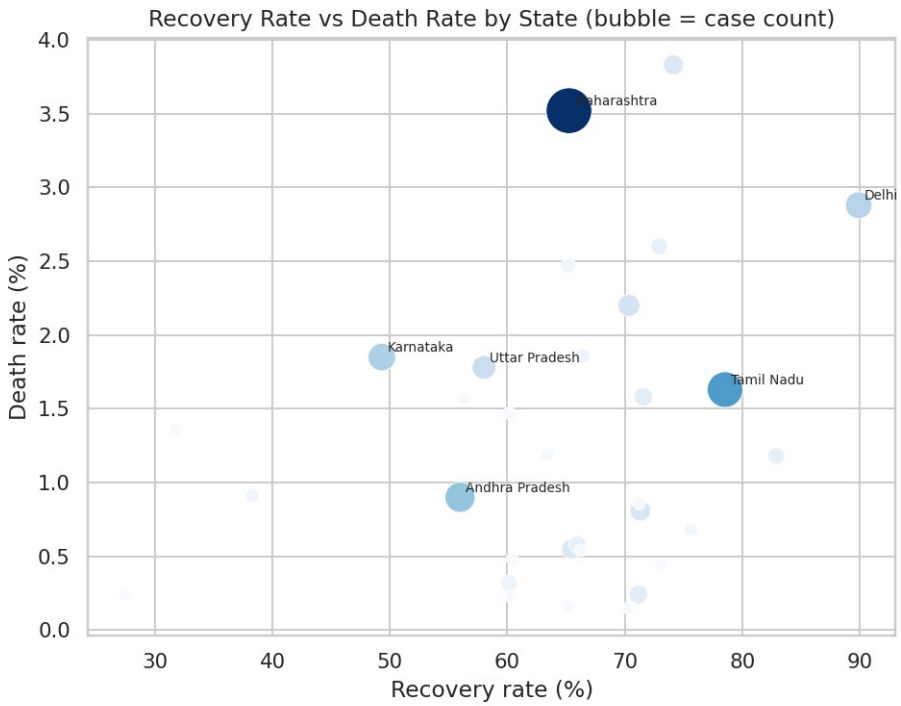


Figure 8: Recovery rate vs death rate by state (bubble size = case count).

Finding

Among the worst-hit states, Gujarat had the highest death rate, at nearly 3.8% — noticeably above the 2.1% national average — while lower-case states like Bihar and Telangana had death rates under 1%. This kind of gap is worth digging into further: differences in health infrastructure, age demographics, and testing intensity could all be contributing factors, though this dataset alone can't tell us which.

3.4 How metrics relate to each other, and how the outbreak evolved month by month

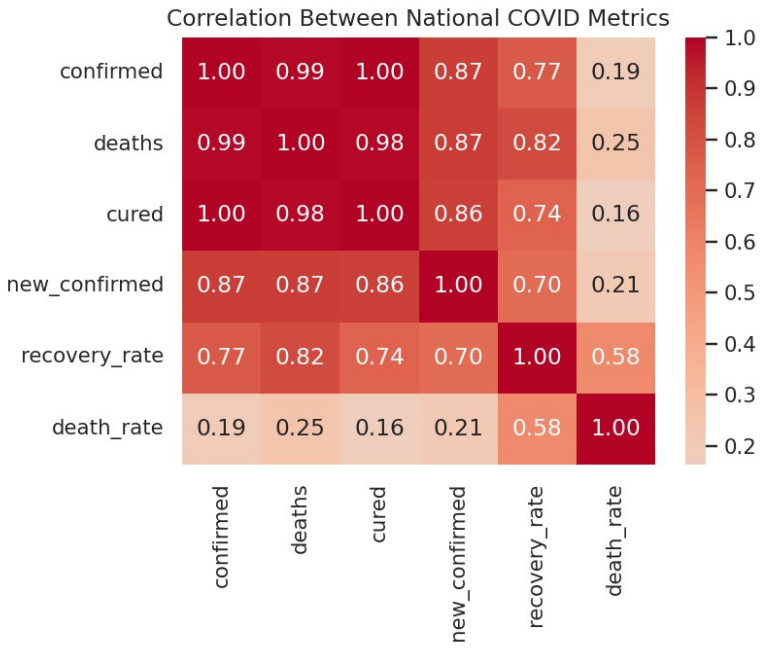


Figure 9: Correlation between national COVID-19 metrics.

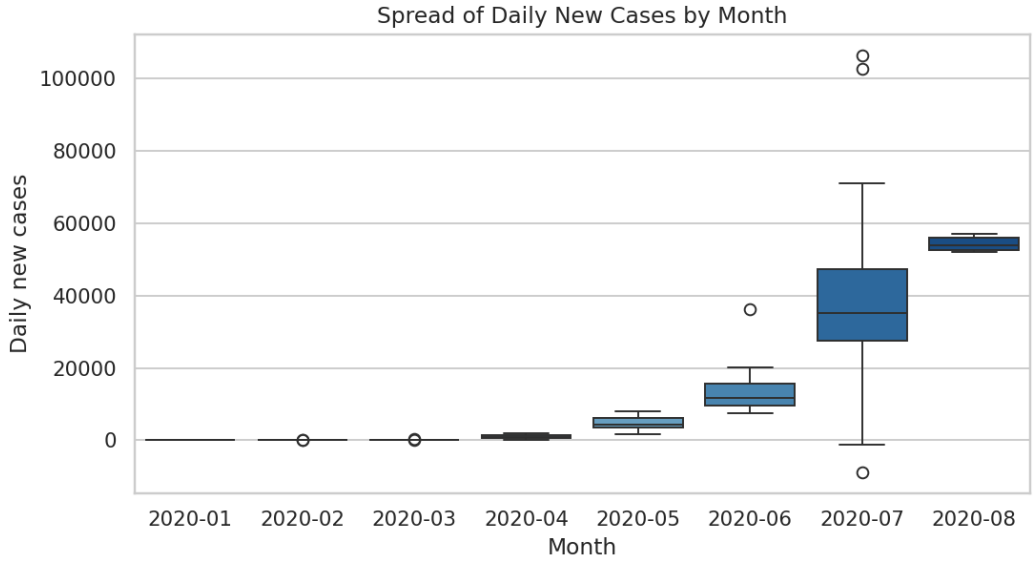


Figure 10: Spread of daily new cases, by month.

Finding

The monthly boxplot makes the acceleration obvious: both the median daily case count and its spread grew every month from February to August 2020, with July and August showing the widest range of daily new cases — a visual confirmation that the outbreak's pace, and its day-to-day volatility, both increased over the period covered.

3.5 The other end of the spectrum, and the vaccination follow-up

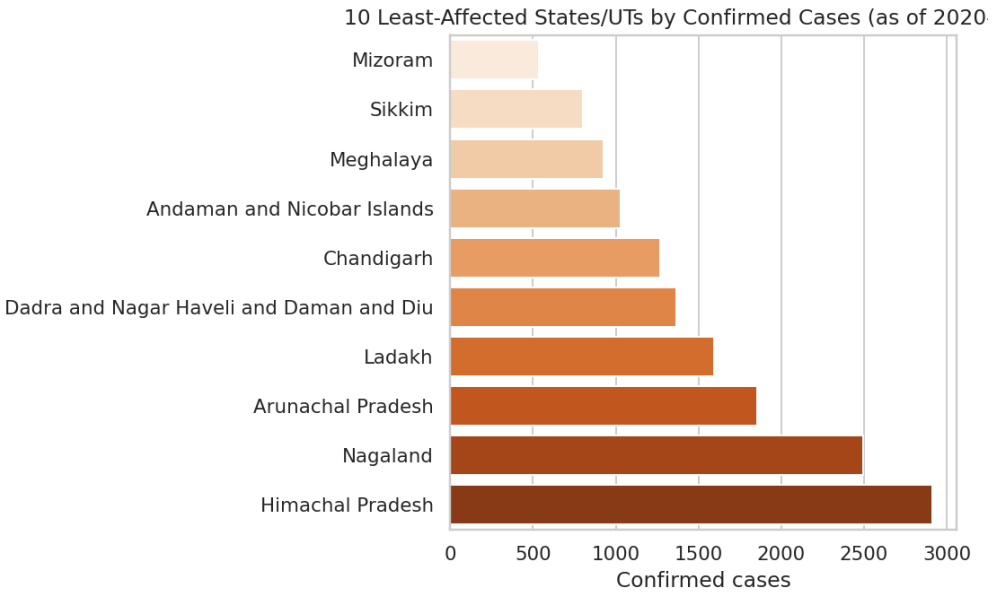


Figure 11: The ten least-affected states/UTs by confirmed case count.

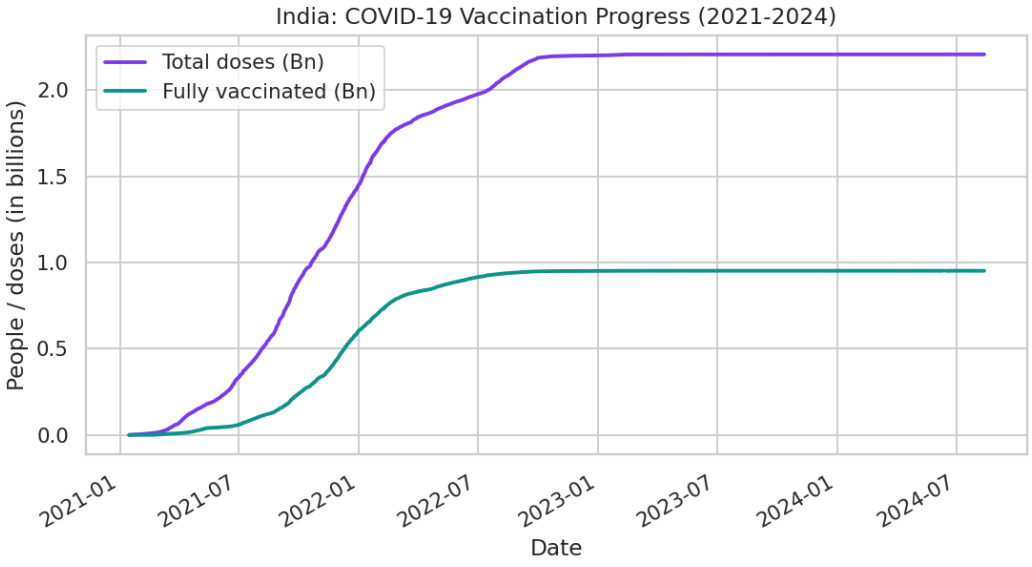


Figure 12: India's national vaccination progress, 2021–2024 (OWID data).

Finding

Small northeastern states and union territories — Mizoram, Sikkim, Meghalaya, and the Andaman & Nicobar Islands — recorded the fewest cases, each under 1,100 total confirmed cases by August 2020, likely reflecting both smaller populations and lower population density/connectivity.

Looking ahead to the vaccination data: by mid-2024, India had administered over 2.2 billion vaccine doses and fully vaccinated more than 950 million people — illustrating the scale of the public health response that followed the wave captured in this project's case data.

4 Overall Summary

- The dataset covers 35 states/UTs (after cleaning up duplicate name spellings) across 186 days, from 30 January to 6 August 2020.
- India's first wave was still accelerating at the end of this window — confirmed cases were growing, and the daily new-case count had not yet peaked.
- The outbreak's impact was far from uniform across states: a handful of states carried the large majority of the case load, while several smaller states/UTs stayed in the low thousands.
- Recovery and death rates both varied meaningfully by state, hinting at real differences in health-care capacity, testing, and demographics worth investigating with additional data.
- The vaccination rollout that followed (2021 onward) reached over 2 billion doses administered by 2024 — a useful reminder that this case dataset captures only the opening chapter of India's COVID-19 response.

4.1 Caveats

This is state-bulletin-sourced data, so it inherits whatever inconsistencies existed in how individual states reported their numbers (hence the spelling and text-formatted Death column issues fixed in Step 3). Recovery and death rates here are simple ratios of *cumulative* totals on a single date, not adjusted for reporting lag or right-censoring, so they should be read as directional trends rather than precise clinical outcome rates.

4.2 Suggestions for further analysis

- Bring in state population figures to convert raw case counts into per-capita rates for a fairer state-to-state comparison.
- Join district-level data (also available in the same source repository) to see which *cities*, not just states, drove the national numbers.
- Extend the case time series past August 2020 (using a more recent source) to see the full shape of India's first and second waves side by side.